Active-layer thickness prediction and regional differences across permafrost regions using a physics-residual machine-learning workflow

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Abstract

Active-layer thickness (ALT) changes from year to year, but most 1 km ALT products are fitted to multi-year conditions and are rarely evaluated against observations of the same cell and year. We assembled 19,172 cell-year ALT observations (2010–2024) from Alaska, the Lena Delta and north-western Canada, paired each with ERA5-Land thawing degree-days of the observation year accumulated to the observation day, and compared the Stefan model, direct machine learning (ML) and a physics-residual workflow in which gradient boosting learns the residual of a recalibrated Stefan model. Under spatial-block cross-validation, the physics-residual model reduced the Stefan error by 3.6 cm in the Lena Delta and 3.7 cm in Canada and was on par in Alaska. When a region was withheld from training, the direct-ML error in the Lena Delta increased by 17.8 cm, whereas the physics-residual model stayed close to Stefan. In the Lena Delta, 25 labels from the region restored accuracy, and five labels lowered the error in external permafrost-monitoring regions from 56.2 to 40.3 cm. In the same cells and years, the Lena Delta error (21.1 cm) was lower than that of five published products (28.4–34.6 cm). The workflow yields year-labelled 1 km maps for 2011–2024.

Keywords: permafrost, active-layer thickness, Stefan model, physics-guided machine learning, spatial cross-validation, ERA5-Land

Introduction

The active layer is the ground above permafrost that thaws in summer and refreezes in winter. Its thickness (ALT) controls the depth of seasonal thaw, the stability of infrastructure and the amount of soil carbon exposed to decomposition[1–3]. ALT changes from year to year with the summer heat input, and the year-to-year signal is part of what monitoring programmes such as the Circumpolar Active Layer Monitoring network record[4,5]. Observations, however, are sparse and uneven: most measurements come from a few dense campaigns and long-term grids, and many sites are visited only once[6,7].

Several 1 km ALT products cover the northern permafrost region. Process-based estimates such as ESA CCI Permafrost run a heat-conduction model with remotely sensed land-surface temperature[8,9]. Statistical and machine-learning (ML) products relate site observations to climate, terrain and soil covariates[10–13]. These products are typically trained on multi-year mean conditions or on site observations from many years pooled together, and they are evaluated with different site sets and validation schemes. A product value therefore cannot be compared directly with an observation of a given cell and year, and products differ by tens of centimetres in the same cell. Validation design adds a further uncertainty: random cross-validation of spatially clustered observations overstates the accuracy of ML models for new locations[14–17].

Physical models offer a complementary route. The Stefan solution, ALT $= E\sqrt{\mathrm{TDD}}$, links thaw depth to the thawing degree-days (TDD) of the season through a single coefficient E that depends on soil thermal properties and moisture[18,19]. It reproduces the seasonal course of thaw and responds to the climate of each year, but one coefficient per region leaves systematic residuals that vary with soil organic carbon and terrain. Combining physical models with ML has improved environmental predictions when the physical model carries the dominant response and the ML model learns what the physics misses[20]. For ALT, process-based and ML estimates have been combined for Alaska[21], but it is not known how such a combination behaves when the input climate is matched to the observation year, when it is transferred to a region without labels, or how many labels a new region needs.

Here we address these questions with a cell-year design: every ALT observation is a 1 km cell in a given year, and every climate input is computed for that year from ERA5-Land, accumulated to the observation day[22]. Using 19,172 cell-year observations from 2010–2024 in Alaska, the Lena Delta and north-western Canada, we compare the Stefan model, direct ML and a physics-residual workflow in which gradient boosting learns the residual of a recalibrated Stefan model. The models are evaluated under five validation designs, along label-number ladders within and across regions, in external permafrost-monitoring regions and against six published products in the same cells and years. We report (1) the effect of the year-specific climate input, (2) the gain of the physics-residual model over the Stefan model and its conditions, (3) the number of labels needed in a new region and (4) the differences from published products, and we produce year-labelled 1 km maps for 2011–2024.

Results

Observations and the year-specific climate input

The label set contains 15,540 cell-years in Alaska (13,253 cells, 2010–2024), 2,436 in the Lena Delta (2,206 cells, 2010–2022) and 1,196 in north-western Canada (730 cells, 2010–2019 and 2021–2024) (Fig. 1a,b and Table 1). The years are unevenly covered: 51% of the Alaska cell-years were observed in 2014, whereas the Lena Delta labels are spread over 13 years. Most observations were made in August (median day of year 224 in Alaska, 234 in the Lena Delta and 237 in Canada), before the end of the thaw season, and thaw continues after the observation day (Supplementary Fig. S1). The thawing degree-days accumulated to the observation day were therefore smaller than the full-season sum, with median ratios of 0.73, 0.81 and 0.89 (Fig. 1d). All models in this study use the climate of the observation year from ERA5-Land monthly data for 2010–2024, accumulated to the observation day; the comparison input is the 2011–2024 mean of the same variables, which corresponds to the multi-year inputs of existing products (Fig. 1c,d). Fig. 2 summarises the physics-residual workflow, the four model structures compared and the five validation designs.

Replacing the mean climate input by the year-specific input reduced the spatial-block error in the Lena Delta for the recalibrated Stefan model (−1.4 [−2.6, −0.8] cm) and for the physics-residual model (−1.6 [−2.4, −1.0] cm), and in Canada for the physics-residual model (−1.5 [−2.0, −0.4] cm) (Fig. 3a). In Alaska the change was not resolved (Stefan +1.8 [−0.1, +3.9] cm), because most labels come from a single year and the year-specific input then adds little information beyond the spatial pattern. The year-specific input is also what makes year-labelled maps possible: every prediction refers to the climate of a stated year.

Validation design and transfer to a withheld region

The ranking of methods depended on the validation design (Fig. 3b). Random cross-validation by cell gave the lowest errors for direct ML (Alaska 11.8 cm), and spatial-block cross-validation with 0.5° blocks raised the direct-ML error by 5.4, 3.6 and 9.9 cm in Alaska, the Lena Delta and Canada. Holding out whole years raised the direct-ML error in the Lena Delta from 15.9 to 20.2 cm, because its labels are spread over many years, and holding out blocks and years together gave the highest within-region errors of all methods in all three regions. When a whole region was withheld and the models were trained on the other two regions, the direct-ML error in the Lena Delta rose to 39.3 cm, compared with 21.5 cm for the Stefan model with the coefficient of the other regions (difference +17.8 [+8.0, +23.9] cm). The physics-residual model stayed close to Stefan in the same test (+2.0 [−0.8, +3.4] cm) and was better than Stefan in Canada (−3.5 [−5.1, −0.1] cm). Spatial-block cross-validation is therefore used as the main design below, and region hold-out as the test of transfer.

Gain of the physics-residual model and its conditions

Under spatial-block cross-validation with the year-specific input (2010–2024 labels), the physics-residual model reduced the error of the recalibrated Stefan model from

21.0 to 17.4 cm in the Lena Delta (−3.6 [−7.8, −0.8] cm) and from 30.6 to 26.9 cm in Canada (−3.7 [−4.8, −0.8] cm). In Alaska both models reached about 18.4 cm (−0.2 [−0.6, +0.1] cm). The gain required terrain and soil covariates (Fig. 3c). With climate covariates only, the residual model did not improve on Stefan in Alaska (+5.9 [−0.3, +14.2] cm) and Canada (+3.9 [−5.1, +10.3] cm); adding terrain and soil gave the gains above, whereas adding MODIS vegetation and snow timing (2010–2024) or high-resolution soil and land-cover classes changed the result by less than 0.5 cm in the Lena Delta and Canada; in Alaska the high-resolution soil and land-cover classes added a small gain (−0.8 [−1.4, −0.2] cm).

The position of the physical information mattered (Supplementary Fig. S2). Adding pseudo-labels from the Stefan model to the training set reproduced the Stefan model (Lena Delta −0.3 [−0.4, −0.1] cm), and using the Stefan prediction as an input gave a result similar to direct ML. Within a region with all labels, direct ML reached a lower point error than the physics-residual model in the Lena Delta (15.9 against 17.4 cm), but its interval against Stefan included zero there and in Alaska, and it failed under region hold-out (above). Replacing CatBoost by random forest, a multilayer perceptron, TabM or FT-Transformer did not change the sign of the physics-residual gain in the Lena Delta and Canada.

Within regional terciles of eight covariates the gain appeared in most conditions of the Lena Delta (19 of 23 terciles with an interval excluding zero) and Canada (17 of 23), and it was largest where the Stefan model had the largest error: across terciles the correlation between the Stefan error and the gain was −0.99 in the Lena Delta and −0.79 in Canada (Fig. 3d). In the warmest TDD tercile of the Lena Delta the gain was −6.2 [−12.0, −0.7] cm. In Alaska no tercile changed by more than 0.9 cm. The residual model therefore acts as a correction of the Stefan coefficient where soil and terrain make the coefficient deviate from its regional value, which is also where its inputs carry the most weight: terrain contributed 50% of the mean absolute SHAP value of the residual model in the Lena Delta and soil 39% in Canada (Supplementary Fig. S3).

Few labels and new regions

The label ladders show how the methods behave when labels are scarce (Fig. 4a). With 25 training labels drawn from the 2010–2024 cell-years of a region, the physics-residual model was at least as accurate as a Stefan model recalibrated on the same labels (Alaska −0.9 [−1.9, +0.1], Lena Delta −0.8 [−2.2, −0.1], Canada −0.3 [−3.2, +1.7] cm), and with 100 labels its gain was resolved in all three regions (−0.7 [−1.4, −0.0], −2.0 [−5.3, −0.3] and −1.9 [−3.7, −0.3] cm). Direct ML with 25 labels had a higher point error than Stefan in Alaska (+2.8 [−0.3, +4.5] cm) and Canada (+1.2 [−1.5, +3.4] cm).

In the transfer experiment a region was predicted by models trained on the two other regions plus n labels from the target region (Fig. 4b). Without target labels, direct ML gave 39.5 cm in the Lena Delta, against 21.5 cm for Stefan with the source coefficient. With 25 target labels the error fell to 19.9 cm for the physics-residual model and 21.3 cm for direct ML. In Alaska the physics-residual model gave the lowest error at every n (17.8 cm without target labels). In Canada direct ML trained on the source

regions was the most accurate (24.2 cm), and recalibrating the Stefan coefficient on 25 Canadian labels increased the error from 27.7 to 31.3 cm, and the physics-residual model, which uses this anchor, followed it (24.3 to 26.3 cm), reflecting the wide range of thaw depths among the Canadian sites.

The same procedure was applied to eight external CALM regions (73 cells, 2010–2024 observations) that were not used for model development (Fig. 4c and Supplementary Fig. S4). Without local labels the physics-residual model trained on the three study regions gave 56.2 cm on the 474 cell-years of seven regions that also have values of Wei et al., Liu et al. and ESA CCI v5. Five labels from the same external region reduced the error to 40.3 cm and 25 labels to 39.7 cm, whereas direct ML reached 47.5 cm with 25 labels. With five labels the physics-residual model was below the ESA CCI v5 product, which was not trained on ALT observations (60.7 cm; difference −19.5 [−31.3, −9.2] cm). Wei et al. and Liu et al. reached 18.0 and 23.2 cm at these sites, which are part of their training data. Most of the gain came from recalibrating the Stefan coefficient; the residual model added no further reduction at these sites (+0.4 [−0.5, +1.2] cm with five labels), where the thaw depths of West and Central Siberia and Svalbard exceed the range of the study regions.

Comparison with published products in the same cells and years

Products and models were compared on the cell-years at which Wei et al., Liu et al. and ESA CCI v4 all provide a value of the observation year, which restricts the comparison to 2010–2020 (Fig. 5a). In the Lena Delta the physics-residual model gave 21.1 cm, lower than ESA CCI v5 (28.4 cm), Aalto et al. (30.6 cm), ESA CCI v4 (31.0 cm), Liu et al. (33.0 cm) and Wei et al. (34.6 cm). Away from product training sites (more than 25 km), the paired differences were −12.5 [−15.6, −5.8] cm against Wei et al. and −5.0 [−9.5, −2.5] cm against ESA CCI v5. In Alaska (17.1 against 17.9 and 18.8 cm) and Canada (25.8 against 27.0 and 27.9 cm) the model and the two ML products were on par.

The difference between regions follows the training data of the products. The Alaska label cells lie at a median distance of 7 km from the training sites of Wei et al., and Liu et al. were trained on an earlier version of the same Alaska and Canada field data. The Lena Delta cells lie at a median distance of 107 km (Supplementary Fig. S5), and both products overestimate the thin active layers of the northern delta (mean bias +21.1 and +19.9 cm north of 72°N). A model trained without any Lena Delta label was already more accurate there than Wei et al. on the same cell-years (26.0 against 30.3 cm; −4.3 [−6.1, −0.8] cm).

Year by year, the physics-residual model had a lower error than Wei et al. and than Stefan in all 6 Lena Delta years with at least 30 cell-years (2013–2017 and 2020), in 7 of 13 Alaska years against Wei et al. and 9 of 13 against Stefan (2010–2022), and in 7 of 8 Canadian years against Stefan (Fig. 5b). At Alaska cells observed in several years, the correlation between observed and predicted year-to-year changes was 0.28 for the physics-residual model and 0.15 and 0.12 for Wei et al. and Liu et al.; in the Lena Delta the products followed the year-to-year changes more closely (0.55 and 0.42 against 0.29).

The residual model can also be placed on top of a product. With the Lena Delta labels, a residual learner added to the linearly recalibrated product of Wei et al. changed its error by −2.3 [−7.4, −0.4] cm, and the Stefan-anchored model remained more accurate than this product-anchored model (−2.8 [−6.2, −0.1] cm; Supplementary Fig. S6). In Alaska and Canada the two anchors were not distinguishable.

Year-labelled 1 km maps

The physics-residual models fitted on all cell-years of Alaska and of the Lena Delta were applied to 1 km grids with the ERA5-Land climate of each year from 2011 to 2024 (Fig. 6). The grids cover cells with an ESA CCI permafrost probability of at least 50% (858,517 cells in Alaska and 38,086 in the Lena Delta). The regional mean anomaly relative to the 2011–2024 mean ranged from −3.2 cm in 2012 to +7.1 cm in 2019 in Alaska and from −9.4 cm in 2017 to +10.0 cm in 2020 in the Lena Delta, with different spatial patterns in each year (Fig. 6b). In the Lena Delta the mean anomaly of the observed cells had the same sign as the predicted regional anomaly in 8 of 10 years with at least 50 cell-years (2012–2021; $r = 0.75$). In Alaska, where the observed cells change from campaign to campaign, the agreement was weaker ($r = 0.25$) and the predicted interannual range (10.2 cm) was smaller than that of the observed cells (17.7 cm) (Fig. 6c). 56% of the Alaska cells and 62% of the Lena Delta cells are inside the area of applicability, and the half-width of the 90% prediction interval is 25.5 and 24.5 cm. Cells outside the area of applicability, mostly mountains and the southern margin of the permafrost zone, are marked in the maps (Supplementary Fig. S7). Supplementary Fig. S8 compares our maps with those of Wei et al. and Liu et al. for Alaska in 2014 and the Lena Delta in 2016, and Supplementary Fig. S9 shows the correction that the residual model adds to the Stefan model.

Discussion

This study evaluated ALT models in a cell-year design, in which the labels, the climate input and the scores refer to the same cell and year. Three results follow. First, with the climate of the observation year accumulated to the observation day, a gradient-boosting model of the Stefan residual reduced the Stefan error by 3.6 and 3.7 cm in the Lena Delta and Canada and matched it in Alaska. Second, the same model remained stable when a region was withheld or when only a few labels were available, whereas direct ML failed in the withheld Lena Delta and needed more labels. Third, in the same cells and years the model was more accurate than five published products in the Lena Delta and on par with the two ML products in Alaska and Canada.

The physics-residual structure works because the two parts carry different information. The Stefan term carries the response to the thaw-season climate of each year and extrapolates to new regions and years with a physically bounded form. The residual model corrects the deviation of the regional coefficient that comes from soil and terrain: the gain needed terrain and soil covariates, it was largest in the conditions where the Stefan error was largest, and the attribution of the residual model pointed

to terrain in the Lena Delta and soil organic carbon in Canada. Adding Stefan pseudo-labels pulled the learner towards the Stefan model and did not add accuracy, and direct ML, which has to learn the climate response from the labels, lost 17.8 cm of accuracy in the withheld Lena Delta. With all labels of a region, direct ML can reach a similar or lower point error than the residual model; the advantage of the residual structure is that its accuracy does not depend on having many labels from the same region.

The published accuracies of existing products refer to different site sets and validation schemes (for example a random ten-fold RMSE of 21.6 cm for Liu et al.[12] and a leave-one-site-out RMSE of 60.5 cm for Wei et al.[13]) and cannot be compared with each other; scoring models and products on the same cells and years removes this ambiguity. The regional differences in this comparison follow the training data of the products. Wei et al. and Liu et al. were trained on CALM, GTN-P and ABoVE observations that lie within a few kilometres of the Alaska label cells, and in Alaska and Canada they were as accurate as our model. The Lena Delta lies about 107 km from the nearest training site of Wei et al., and both products overestimate its thin active layers. A model trained without any Lena Delta label was already more accurate there than Wei et al., and with the Lena Delta labels the error fell to 21.1 cm, against 28.4–34.6 cm for the products. The residual learner also improved a product used as the anchor. Local labels combined with residual learning can therefore be used to adapt either the Stefan model or an existing product to a region in which the product was not trained.

The label experiments give a practical sequence for a new region. Without local labels, the Stefan model with the coefficient of other regions is the safer starting point, because direct ML can fail. With about 25 labels, recalibrating the Stefan coefficient and learning the residual restored the accuracy in the Lena Delta, and in the external CALM regions five labels reduced the error from 56.2 to 40.3 cm, below the ESA CCI v5 product. The external sites also show the limit of the transfer: West and Central Siberia and Svalbard have thaw depths beyond the range of the study regions, and the gain there came from recalibrating the coefficient rather than from the residual model. Canada shows the opposite case, in which the source-trained direct ML was more accurate and recalibration on a small Canadian sample increased the error. The number of local labels needed therefore depends on how far the new region lies from the training range, and 25–100 labels covering the range of soils and terrain are a reasonable target for regions similar to the study regions.

The year-labelled maps are the second product of the design. Each map is driven by the ERA5-Land climate of a stated year, so that a map can be compared with observations or other products of the same year and anomalies can be followed from year to year (for example +7.1 cm in Alaska in 2019). The maps report the area of applicability and a 90% prediction interval, which mark the cells where the prediction leaves the training conditions. The predicted interannual range is smaller than that of the observed cell means (see also the long-record cells in Supplementary Fig. S10), partly because the observed cells change between campaigns and partly because the residual model is trained mainly on spatial contrasts.

Several limits apply. Half of the Alaska labels come from 2014, so the effect of the year-specific input could not be resolved there. The Lena Delta labels come from one dataset in 16 blocks, which widens the intervals. The climate input is monthly and the soil covariates are static, so short events and changes in soil moisture are not represented. Active layers thicker than about 3 m, mostly in mountains and at the southern margin of the permafrost zone, lie outside the training range. Daily climate inputs, snow and soil-moisture information and labels from a fourth region such as West Siberia, where both CALM and campaign data exist, are the next steps.

In summary, a Stefan model recalibrated with local labels and corrected by a residual learner gives year-specific ALT that is more accurate than the Stefan model where soil and terrain matter, stable in regions without labels and adaptable with a few labels. Evaluated in the same cells and years, it is more accurate than published products in a region they did not learn and on par where they did. The same procedure can be applied to other permafrost regions and to existing products as anchors, and it provides year-labelled 1 km maps with a stated area of applicability for monitoring the active layer.

Methods

Study regions and ALT labels. ALT observations were taken from three public sources. For Alaska and north-western Canada we used the ABoVE soil moisture and active-layer thickness compilation, version 2[6], restricted to probe measurements made in August–October with $0 < \mathrm{ALT} < 300$ cm (measurements in January–July were excluded as early-season thaw depths), and the CALM end-of-season grid values[5]. For the Lena Delta we used the ALLena thaw-depth data[7], restricted to August–September measurements without the 'exceeds probe length' flag, taking the seasonal maximum per location and year. Observations were assigned to 1 km cells and averaged within each cell and year, and the mean day of year of the measurements was kept; CALM values without a date were assigned day 244 (1 September). Only years with yearly climate input (2010–2024) were used. The resulting label set has 15,540 cell-years in Alaska (2010–2024), 2,436 in the Lena Delta (2010–2022) and 1,196 in Canada (2010–2019 and 2021–2024), in 73, 16 and 26 blocks of 0.5° (Table 1). 203, 25 and 48 cells have observations in four or more years.

Year-specific climate input. Monthly ERA5-Land data for January 2010 to December 2024[22] (2 m air temperature, soil temperature of the top layer and snow water equivalent) were taken at the nearest land grid cell of each 1 km cell (fallback up to 0.5° for coastal cells). For each cell and year y we computed eight climate variables: mean annual air temperature, TDD (sum of positive monthly mean temperatures times days), $\sqrt{\mathrm{TDD}}$, the warmest monthly mean, the freezing degree-days and the coldest monthly mean of the preceding winter (October $y-1$ to April y), the mean top-layer soil temperature and the mean snow water equivalent of the hydrological year (October $y-1$ to September y). For 2010 the winter variables use January–April only. The main input (year-specific climate input) accumulates TDD to the observation day by linear interpolation of the monthly cumulative TDD of the observation year. Two alternatives were run on the same cell-years: the full-season TDD

of the observation year, and the 2011–2024 mean of the yearly values (mean climate input), which corresponds to the multi-year inputs of existing products. Maps use the full-season TDD of each year (end-of-season ALT).

Static covariates. Terrain variables (elevation, slope, the sine and cosine of aspect, topographic position index and roughness) were computed from the 30 m Copernicus DEM[23] in a 33 × 33 pixel window (about 1 km) around each cell. Soil variables (clay, sand, silt, bulk density, coarse fragments and pH at 5–15 cm, and soil organic carbon at 0–5, 5–15 and 15–30 cm) were sampled from SoilGrids 2.0 at 250 m[24]. The main feature set (23 variables) contains the eight climate variables, six terrain variables and nine soil variables. Sensitivity sets were climate only; climate and terrain; the main set plus MODIS summer NDVI and EVI and snow-off timing (MOD13Q1 and MOD10A2 version 6.1 for 2010–2024 via AppEEARS[25,26]); the main set plus high-resolution soil carbon and land-cover classes[27]; and all variables including the climate of the previous year. The ESA CCI permafrost probability (1997–2021 mean)[8] was used only to define the map domain (probability $\geq 50\%$), not as a model input.

Published products. Six products were extracted at the label cells: Wei et al.[13] (1 km, 2000–2024), Liu et al.[12] (1 km, 2003–2020), ESA CCI Permafrost ALT v4 (1997–2021) and v5 (1997–2023)[8,9], Yi and Kimball[28] (Alaska, 2001–2015) and Aalto et al.[10] (baseline 2000–2014, one value per cell). For yearly products the value of the observation year was used; the years available at the label cells were 2010–2024 (Wei et al.), 2010–2020 (Liu et al.), 2010–2021 (CCI v4), 2010–2023 (CCI v5) and 2010–2015 (Yi and Kimball). Products were used only as comparison targets. Wei et al. publish the coordinates of their training sites; the distance from each label cell to the nearest training site was used for the 25 km mask.

Models. The Stefan model writes ALT $= E\sqrt{\mathrm{TDD}}$[18]. The source-coefficient Stefan model takes E fitted by least squares on the other two regions; the recalibrated Stefan model fits E on the training labels of the region. Direct ML predicts ALT from the 23 features with CatBoost gradient boosting[29] (600 iterations, learning rate 0.03, depth 6, L2 regularisation 3). Physics-input ML adds the recalibrated Stefan and Kudryavtsev[30] predictions as features. Physics pseudo-labels add to the training set an equal number of rows drawn from the cells of the scoring fold, with their covariates and the training-fold Stefan prediction as the target (no observed values). The physics-residual model keeps the recalibrated Stefan prediction $\hat{y}_P$ as the anchor and adds a CatBoost model g of the residual:

$$\hat{y} = E\sqrt{\mathrm{TDD}} + \lambda\, g(\mathbf{x}), \qquad g \approx y - \hat{y}_P, \tag{1}$$

with $\lambda = 0.5$ (0.25, 1 and a value chosen by inner block cross-validation were run as sensitivity cases). All learned models were run with three seeds and the predictions averaged. Random forest, a multilayer perceptron, TabM and FT-Transformer[31–33] replaced CatBoost in a learner sensitivity test. SHAP values of the residual and direct models were computed with the CatBoost implementation[34] on models fitted to all labels of each region.

Validation designs. Five designs were applied within each region[14,15]: random five-fold by cell-year; spatial-block five-fold, with 0.5° blocks assigned to folds balanced by the number of cell-years (main design); year hold-out, in which each observation year with at least 20 cell-years is scored by models trained on the other years; block-by-year, in which both the block fold and the year of the scored cell-years are excluded from training; and region hold-out, in which a region is scored by models trained on the other two regions after removing training cells within 100 km of the test region. In region hold-out no target labels are available, so the recalibrated Stefan model equals the source-coefficient model.

Label ladders, transfer and external regions. In the within-region ladder, the training labels of each spatial-block fold were subsampled to $n = 25$, 50, 100, 250, 500, 1000, 2500 or all (three random draws), the Stefan coefficient was recalibrated on the same n labels, and the full test fold was scored (all model structures in Supplementary Fig. S11). In the transfer ladder, each region was scored in its spatial-block folds by models trained on all labels of the other two regions plus $n = 0$, 25, 50, 100, 250 or 500 labels drawn from the training folds of the target region. The Stefan coefficient was refitted on the n target labels ($n \geq 3$), and the residual model was trained on the source rows (with the source-coefficient anchor) and the target rows (with the refitted anchor). For the external test, CALM observations from 2010–2024[5] outside the three study regions and outside mountain permafrost (Switzerland, Kazakhstan) were assigned to 1 km cells. Censored values, measurements flagged as early, burnt, flooded or not reaching the permafrost table, and sites within 100 km of the study labels (Samoylov and Bykovsky in the Lena Delta) were removed, leaving 816 cell-years in 73 cells in eight regions (West, Central and North-East Siberia, European Russia, Chukotka, Kamchatka, Svalbard and Greenland). Climate inputs and covariates were computed in the same way. Each external region was scored block by block (cell by cell for Greenland, which has one block), with models trained on the three study regions plus $n = 0$, 5, 10, 25, 50 or all labels from the other blocks of the same external region. One Svalbard borehole with a mean ALT of about 6 m, outside the training range, was excluded.

Scoring and statistics. Root-mean-square error (RMSE), bias and Pearson correlation were computed on out-of-fold predictions. Differences between two methods (ΔRMSE = RMSE of the first minus RMSE of the second) were computed on the same cell-years, and 95% confidence intervals were taken from 2,000 bootstrap resamples of the 0.5° blocks (1,000 for the ladders). We describe a difference as resolved when its interval excludes zero. Conditions of the gain were examined within regional terciles of eight covariates (full-season TDD of the observation year, freezing degree-days, snow water equivalent, soil organic carbon, clay, slope, elevation and summer NDVI); terciles with fewer than 30 cell-years or fewer than five blocks were not scored. Year-wise RMSE was computed for years with at least 30 cell-years, and the correlation between observed and predicted year-to-year changes was computed at cells observed in several years.

Comparison with products. Products and models were compared on the same cell-years. The common-cell comparison uses only cell-years at which Wei et al., Liu

et al. and ESA CCI v4 all provide a value of the observation year (2010–2020). A leakage mask removes cells within 25 km of the training sites of Wei et al.; Liu et al. were trained on an earlier version of the ABoVE data used here[12], so their Alaska and Canada values are not independent of our labels. In the product-anchored test, the product value of the observation year was linearly recalibrated on the training fold and corrected by the same residual learner ($\lambda = 0.5$) on the same block folds.

Yearly maps, area of applicability and prediction intervals. For Alaska and the Lena Delta the recalibrated Stefan and physics-residual models were fitted on all cell-years of the region and applied to the 1 km cells with an ESA CCI permafrost probability of at least 50% (858,517 and 38,086 cells), with the full-season climate of each year from 2011 to 2024. The area of applicability was defined following Meyer and Pebesma[35]: in the standardised space of the 23 features (with the 2011–2024 mean climate), a map cell is inside when its distance to the nearest training row does not exceed the 95th percentile of the distances from each training row to the nearest training row in another block. The 90% prediction interval was taken from the 5th and 95th percentiles of the spatial-block out-of-fold residuals. For display, the 1 km values were interpolated bilinearly to the 250 m hillshade grid.

Computation. Model fitting was run on cloud CPU nodes with 96 cores (Rescale) with fixed seeds, and data preparation, aggregation and figures on a local server. Supplementary Table S6 lists the runs.

Use of large language models. A large language model (Claude, Anthropic) was used to assist with code development, figure scripting and language editing. All analyses, numbers and text were checked by the authors, and every number in the manuscript is generated from the stored result tables. [Authors to confirm.]

Code availability. The code for data preparation, modelling, evaluation and figures is available at [repository and DOI]. Every number in the text is produced from the result tables by the scripts that generate the manuscript macros.

Data availability. All input data are public. ALT observations: ABoVE soil moisture and active-layer thickness, version 2 (ORNL DAAC, https://doi.org/10.3334/ORNLDAAC/2369), ALLena thaw-depth data (PANGAEA, https://doi.org/10.1594/PANGAEA.973813) and GTN-P CALM (PANGAEA, https://doi.org/10.1594/PANGAEA.972777). Climate: ERA5-Land monthly averaged data (Copernicus Climate Data Store). Covariates: Copernicus DEM GLO-30, SoilGrids 2.0, ESA CCI permafrost probability, MODIS MOD13Q1 and MOD10A2 version 6.1 (via AppEEARS). Products: Wei et al. (https://doi.org/10.5281/zenodo.21667583), Liu et al. (https://doi.org/10.5281/zenodo.10070610), ESA CCI ALT v4 and v5 (CEDA), Aalto et al. (https://zenodo.org/records/5003804) and Yi and Kimball (https://doi.org/10.3334/ORNLDAAC/1760). The cell-year label table, the yearly climate-input table, the static covariate table, the out-of-fold predictions of all methods and the yearly 1 km maps of Alaska and the Lena Delta (2011–2024) will be deposited in a public repository on acceptance [repository and DOI].

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Additional information. **Supplementary information** accompanies this paper (Supplementary Methods, Figures S1–S11 and Tables S1–S6).

Competing interests The authors declare no competing interests.

Figure legends

Figure 1. ALT observations, inputs and the years they cover. (a) ESA CCI permafrost probability (mean of 1997–2021) with the three study regions and the number of 1 km cells with ALT labels in 2010–2024. (b) Number of cell-year labels per observation year, 2010–2024, for each region. (c) Periods of the data: labels of each region, the ERA5-Land monthly data used for the year-specific climate input (2010–2024), the static covariates (no year), and the six published products with their available period (light) and the years used for comparison at the label cells (dark); the shaded band marks the analysis period 2010–2024. (d) Definition of the climate input at one Alaska CALM cell (71.3°N, 156.6°W) in 2014: cumulative thawing degree-days (TDD) from ERA5-Land, the value accumulated to the observation day (year-specific input used for training and scoring; 1 September, the convention for undated CALM end-of-season values), the full-season value of the same year (used for maps) and the 2011–2024 mean (mean climate input).

Figure 2. Physics-residual workflow and validation designs. (a) Structure of the physics-residual model, with the 2014 inputs of Alaska as an example: the year-specific climate variables and the static terrain and soil covariates enter a gradient-boosting model of the residual of the recalibrated Stefan model ($\mathrm{ALT} = E\sqrt{\mathrm{TDD}}$); the observed ALT is used only in training (dashed); the output is the Stefan estimate plus $\lambda = 0.5$ times the residual, with the area of applicability (computed with the 2011–2024 mean climate). (b) The four model structures compared: direct ML, physics-input ML (Stefan and Kudryavtsev predictions as features), physics pseudo-labels (training set augmented with Stefan predictions) and physics plus residual ML. (c) The five validation designs shown on the Lena Delta label cells (labels of 2010–2022): random, spatial block (0.5°), year hold-out, block-by-year and region hold-out (trained on Alaska and Canada); one example fold is shown for each design, and a cell is drawn as test (black) if any of its cell-years is in the test fold.

Figure 3. Year-specific climate input, validation design and the conditions of the residual gain. All panels use the cell-year labels of 2010–2024 and out-of-fold predictions. (a) Spatial-block RMSE with the mean climate input (2011–2024 mean; open markers) and the year-specific climate input (observation year, accumulated to the observation day; filled markers) for the recalibrated Stefan model, the physics-residual model and direct ML; in this panel only, open and filled markers denote the climate input. (b) RMSE under the five validation designs for the Stefan model with the source coefficient, the recalibrated Stefan model, the physics-residual model and direct ML; year hold-out and block-by-year use the years with at least 20 cell-years; under region hold-out the two Stefan models coincide; values beyond the axis are printed. (c) Difference in spatial-block RMSE between the physics-residual model and the recalibrated Stefan model for six covariate sets. (d) The same difference within regional terciles of eight covariates; cells outlined in black have a 95% interval excluding zero, hatched cells have fewer than 30 cell-years or fewer than five blocks, and 0.0 marks a difference smaller than 0.05 cm. In (a), (c) and (d) bars and intervals are 95% block-bootstrap intervals (2,000 resamples of 0.5° blocks); negative differences favour the physics-residual model.

Figure 4. Accuracy with few labels and in new regions. (a) Within-region label ladder: difference in spatial-block RMSE between each method and a Stefan model recalibrated on the same n training labels drawn from the 2010–2024 cell-years (three draws); the band is the 95% block-bootstrap interval of the physics-residual model. (b) Transfer to a new region: RMSE in the spatial-block folds of each region for models trained on the two other regions plus n labels from the target region. (c) External CALM regions (observations of 2010–2024 outside the study regions): RMSE against the number of labels from the same external region added to the training data, on the 474 cell-years of seven regions with values of Wei et al., Liu et al. and ESA CCI v5 (2010–2020); the thick light line is the Stefan model recalibrated on the same labels, which nearly coincides with the physics-residual model; horizontal lines show the products, of which Wei et al. and Liu et al. were trained on these sites and ESA CCI v5 was not trained on ALT observations.

Figure 5. Comparison with published products in the same cells and years. (a) RMSE of the three methods (spatial-block out-of-fold predictions) and five products on the cell-years at which Wei et al., Liu et al. and ESA CCI v4 provide a value of the observation year (2010–2020); filled markers use all common cell-years and open markers only cells more than 25 km from the training sites of Wei et al.; ESA CCI v5 and Aalto et al. are scored on the common cell-years where they also have a value; values beyond the axis are printed. (b) RMSE per observation year on the same cell-years of each year for the physics-residual model, the recalibrated Stefan model, Wei et al. and Liu et al. (to 2020), for years with at least 30 cell-years: Alaska 2010–2022, Lena Delta 2013–2017 and 2020 and Canada 2013, 2017–2019 and 2021–2024; grey bars give the number of cell-years per year (log scale).

Figure 6. Year-labelled 1 km maps. (a) Mean ALT of 2011–2024 from the physics-residual model in Alaska, computed from the yearly maps; each yearly map uses the full-season ERA5-Land climate of that year. Dots mark the label cells; lighter colours mark cells outside the area of applicability; grey land lies outside the map domain (ESA CCI permafrost probability below 50%, water or missing inputs). (b) Anomaly of each year from 2011 to 2024 relative to the 2011–2024 mean, on the same colour range (±15 cm) for all years. (c) Regional mean anomaly of the predicted maps (lines) for Alaska and the Lena Delta, 2011–2024, and the mean anomaly of the observed cells in years with at least 50 cell-years (circles; Alaska 2011–2022, Lena Delta 2012–2021). The 1 km values are interpolated to 250 m for display.

Table 1 Data and years used. Top: ALT labels by region (cell-years of 2010–2024 with year-specific climate input). Bottom: inputs and products, with the period available and the years used in this study. Blocks are the 0.5° cells used for spatial cross-validation and bootstrap; there are no Canadian labels in 2020.

Region	Cell-years	Cells	Blocks	Years	Cells ≥4 yr	Largest year
Alaska	15,540	13,253	73	2010–2024	203	2014 (51%)
Lena Delta	2,436	2,206	16	2010–2022	25	2016 (28%)
Canada (NW)	1,196	730	26	2010–2024	48	2022 (37%)
External CALM	816	73	50	2010–2024	–	–

Data	Resolution	Available	Used	Role
ERA5-Land monthly[22]	0.1°	1950–	2010–2024; mean 2011–2024	Climate input
Copernicus DEM[23]	30 m	static	–	Terrain
SoilGrids 2.0[24]	250 m	static	–	Soil
MODIS MOD13Q1, MOD10A2[25,26]	250–500 m	2000–	2010–2024	Sensitivity test
ESA CCI permafrost probability[8]	1 km	1997–2021	mean	Map domain
Wei et al.[13]	1 km	2000–2024	2010–2024	Comparison
Liu et al.[12]	1 km	2003–2020	2010–2020	Comparison
ESA CCI ALT v4[8]	1 km	1997–2021	2010–2021	Comparison
ESA CCI ALT v5[9]	1 km	1997–2023	2010–2023	Comparison
Yi and Kimball[28]	1 km	2001–2015	2010–2015 (Alaska)	Comparison
Aalto et al.[10]	30″	2000–2014	time-invariant	Comparison

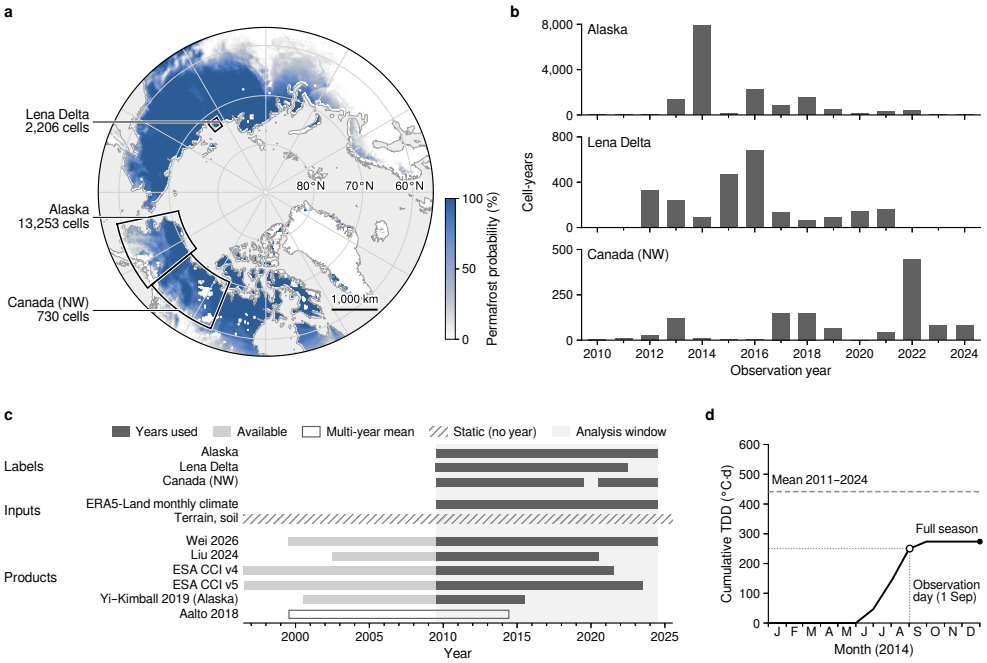

a
Lena Delta
2,206 cells
Alaska
13,253 cells
Canada (NW)
730 cells
80°N
70°N
60°N
1,000 km
100
50
0
Permafrost probability (%)
b
Alaska
Lena Delta
Canada (NW)
Cell-years
8,000
4,000
800
400
500
250
2010
2012
2014
2016
2018
2020
2022
2024
Observation year
c
Years used
Available
Multi-year mean
Static (no year)
Analysis window
Labels
Alaska
Lena Delta
Canada (NW)
Inputs
ERA5-Land monthly climate
Terrain, soil
Products
Wei 2026
Liu 2024
ESA CCI v4
ESA CCI v5
Yi–Kimball 2019 (Alaska)
Aalto 2018
2000
2005
2010
2015
2020
2025
Year
d
Mean 2011–2024
Full season
Observation
day (1 Sep)
Cumulative TDD (°C·d)
600
500
400
300
200
100
0
J F M A M J J A S O N D
Month (2014)

Figure 1

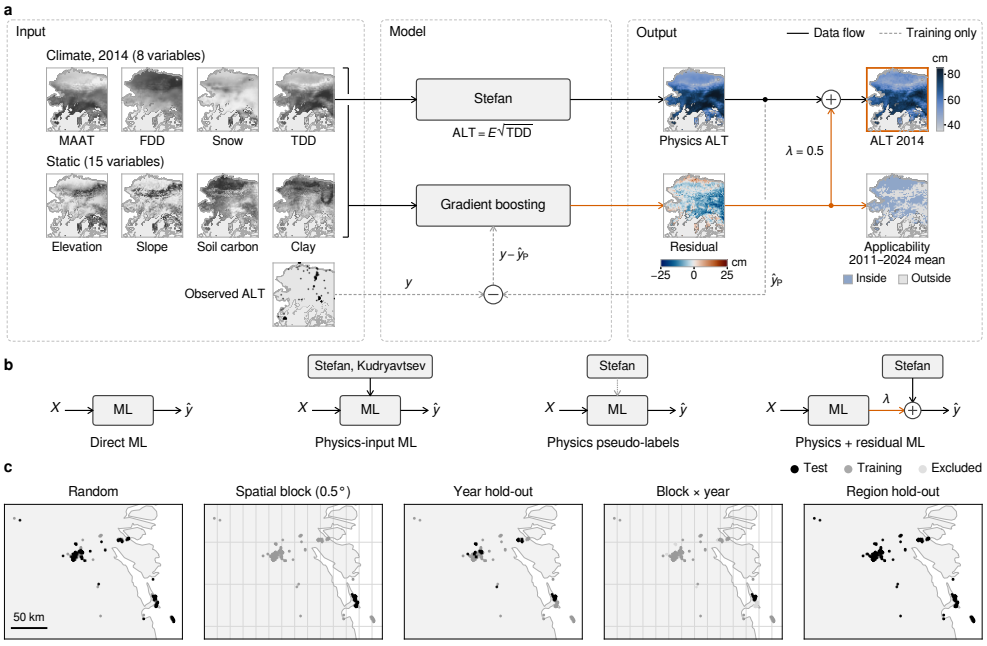

a
Input
Climate, 2014 (8 variables)
MAAT
FDD
Snow
TDD
Static (15 variables)
Elevation
Slope
Soil carbon
Clay
Observed ALT
Model
Stefan
ALT = E√TDD
Gradient boosting
y − ŷP
y
Output
Data flow
Training only
Physics ALT
Residual
−25 0 25 cm
λ = 0.5
ALT 2014
cm
80
60
40
Applicability
2011–2024 mean
Inside
Outside
ŷP
b
X
ML
ŷ
Direct ML
Stefan, Kudryavtsev
Physics-input ML
Stefan
Physics pseudo-labels
Stefan
λ
Physics + residual ML
c
Test
Training
Excluded
Random
Spatial block (0.5°)
Year hold-out
Block × year
Region hold-out
50 km

Figure 2

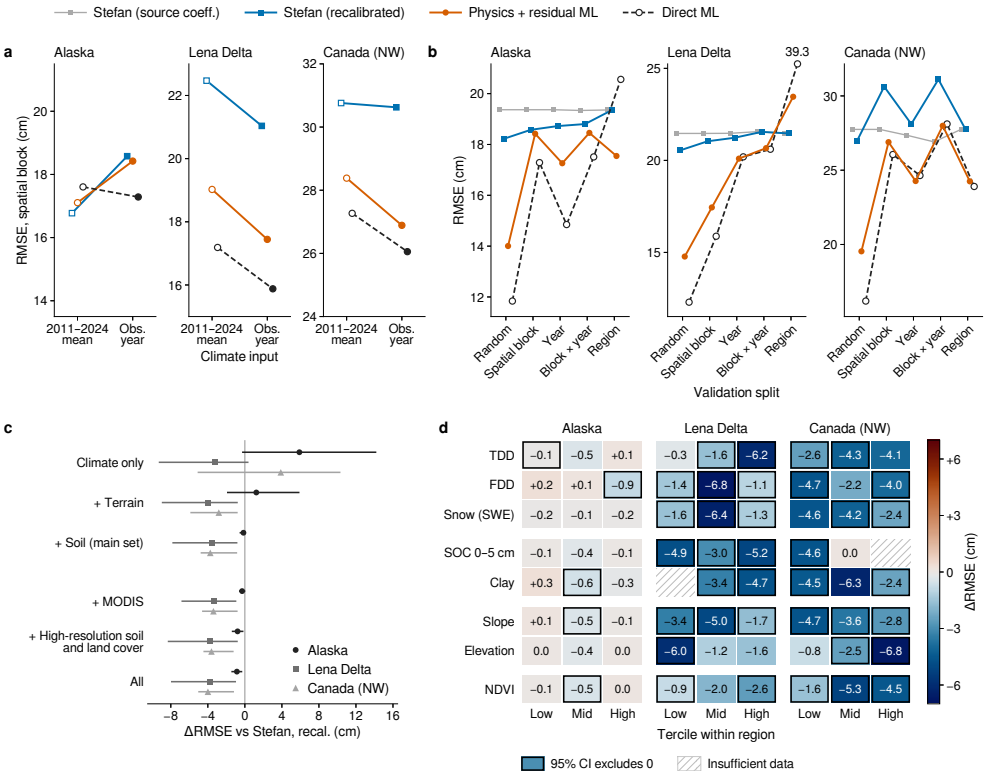

Stefan (source coeff.)
Stefan (recalibrated)
Physics + residual ML
Direct ML
a
Alaska
Lena Delta
Canada (NW)
RMSE, spatial block (cm)
2011–2024 mean
Obs. year
Climate input
b
RMSE (cm)
39.3
Random
Spatial block
Year
Block × year
Region
Validation split
c
Climate only
+ Terrain
+ Soil (main set)
+ MODIS
+ High-resolution soil and land cover
All
ΔRMSE vs Stefan, recal. (cm)
d
TDD
FDD
Snow (SWE)
SOC 0–5 cm
Clay
Slope
Elevation
NDVI
Low
Mid
High
Tercile within region
ΔRMSE (cm)
95% CI excludes 0
Insufficient data

Figure 3

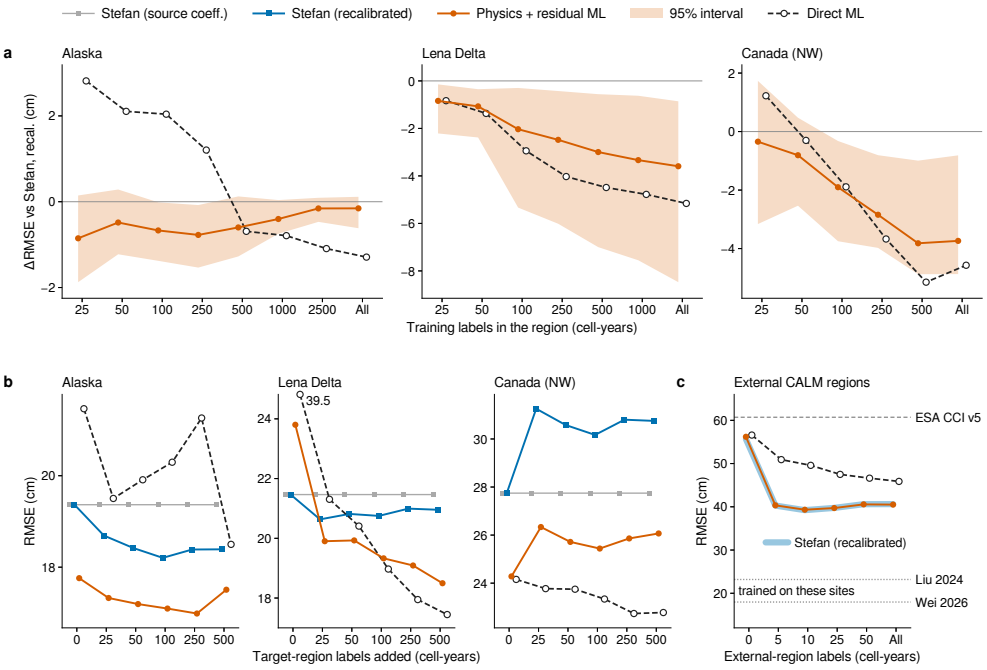

Stefan (source coeff.)
Stefan (recalibrated)
Physics + residual ML
95% interval
Direct ML
a
Alaska
Lena Delta
Canada (NW)
ΔRMSE vs Stefan, recal. (cm)
Training labels in the region (cell-years)
b
RMSE (cm)
39.5
Target-region labels added (cell-years)
c
External CALM regions
ESA CCI v5
Stefan (recalibrated)
Liu 2024
trained on these sites
Wei 2026
External-region labels (cell-years)

Figure 4

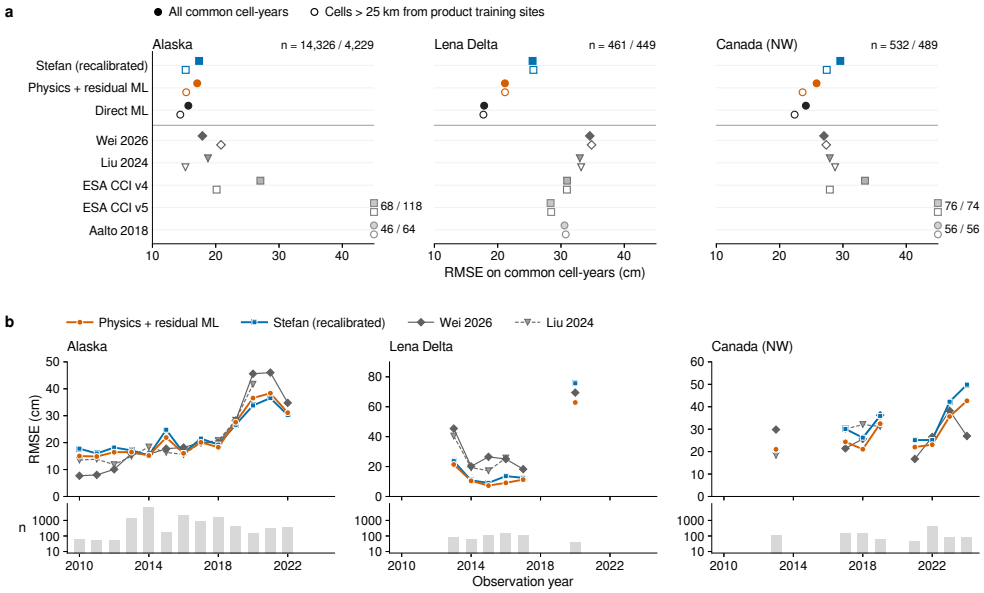

a
All common cell-years
Cells > 25 km from product training sites
Alaska
n = 14,326 / 4,229
Lena Delta
n = 461 / 449
Canada (NW)
n = 532 / 489
Stefan (recalibrated)
Physics + residual ML
Direct ML
Wei 2026
Liu 2024
ESA CCI v4
ESA CCI v5
Aalto 2018
68 / 118
46 / 64
76 / 74
56 / 56
RMSE on common cell-years (cm)
b
Physics + residual ML
Stefan (recalibrated)
Wei 2026
Liu 2024
Alaska
Lena Delta
Canada (NW)
RMSE (cm)
n
Observation year

Figure 5

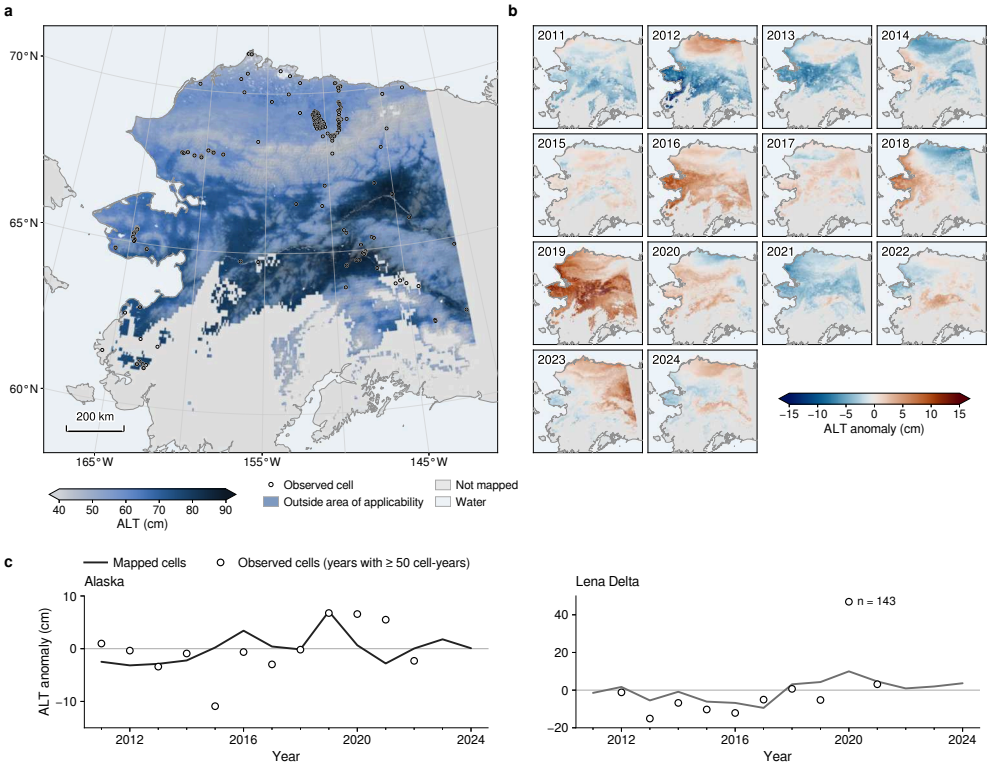

a
70°N
65°N
60°N
200 km
165°W
155°W
145°W
40 50 60 70 80 90
ALT (cm)
Observed cell
Outside area of applicability
Not mapped
Water
b
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
−15 −10 −5 0 5 10 15
ALT anomaly (cm)
c
Mapped cells
Observed cells (years with ≥ 50 cell-years)
Alaska
ALT anomaly (cm)
10
0
−10
2012
2016
2020
2024
Year
Lena Delta
n = 143
40
20
0
−20
2012
2016
2020
2024
Year

Figure 6